Module 2: Embedding Extraction

Project	Face Verification Pipeline -- AgeDB-30 / LFW
Notebook	`notebooks/02_extract_embeddings.ipynb`
Date	2026-03-16
Dependency	Module 1 -- `notebooks/01_detect_align.ipynb`
Input	`data/aligned/{agedb,lfw}/*.jpg` (112x112 aligned crops)
Output	`results/embeddings_{model}_{dataset}.npz`, 7 figures PNG

Objective

Generate L2-normalized 512-dimensional feature vectors for all aligned face crops from the verification pair protocol. Two pretrained models are evaluated:

1. **ArcFace / w600k_r50** (InsightFace, WebFace600K) -- ResNet-50 backbone trained with ArcFace loss. Used in EDA (Module 0) and established as the primary baseline.
2. **GhostFaceNetV2** (DeepFace) -- lightweight GhostNet-V2 backbone trained with ArcFace loss on a large-scale face dataset. Expected EER ~3.0% (AgeDB), ~0.35% (LFW) per project analysis.

Critical design decision: Both models receive pre-aligned 112x112 crops directly, bypassing any internal face detection. For InsightFace, this means calling `get_feat()` directly. For DeepFace/GhostFaceNet, this means `detector_backend='skip'`. Without this, scene-level detectors achieve < 1% recall on face-filling 112x112 crops, corrupting all downstream metrics.

Known failure mode: GhostFaceNetV2 called via raw `model.forward()` without DeepFace's internal preprocessing returns EER ~52%, rendering results meaningless. DeepFace handles normalization correctly -- this is the key value of the framework.

1. Configuration and Imports

```
import os
import time
import hashlib
import shutil
import warnings
from pathlib import Path

import cv2
```

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from scipy import stats
from sklearn.metrics import roc_curve, auc as sklearn_auc
from sklearn.manifold import TSNE
from tqdm.auto import tqdm

# Suppress TensorFlow verbosity
os.environ["TF_CPP_MIN_LOG_LEVEL"] = "3"
os.environ["TF_ENABLE_ONEDNN_OPTS"] = "0"
warnings.filterwarnings("ignore")

plt.style.use("seaborn-v0_8-whitegrid")
plt.rcParams.update({
    "figure.dpi": 120,
    "savefig.dpi": 150,
    "font.size": 10,
    "axes.titlesize": 11,
    "axes.labelsize": 10,
    "figure.facecolor": "white",
})

SEED = 42
np.random.seed(SEED)

PROJECT_ROOT = Path("..").resolve()
DATA_DIR     = PROJECT_ROOT / "data"
RESULTS_DIR  = PROJECT_ROOT / "results"
LOGS_DIR     = PROJECT_ROOT / "logs"
OUTPUT_DIR   = PROJECT_ROOT / "notebooks" / "embedding_output"
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

DATASETS = ["agedb", "lfw"]
ALIGNED_DIRS = {name: DATA_DIR / "aligned" / name for name in
DATASETS}

IMAGE_SIZE = 112
EMB_DIM    = 512

print(f"Project root  : {PROJECT_ROOT}")
print(f"Results dir   : {RESULTS_DIR}")
print(f"Output dir    : {OUTPUT_DIR}")
```

```
Project root  : /Users/jakub/Desktop/NASK_ZADANIE
Results dir   : /Users/jakub/Desktop/NASK_ZADANIE/results
Output dir    :
/Users/jakub/Desktop/NASK_ZADANIE/notebooks/embedding_output
```

2. Methodology

2.1 Model Architectures

Property	ArcFace (w600k_r50)	GhostFaceNetV2
Backbone	ResNet-50	GhostNet-V2
Loss	ArcFace	ArcFace
Training data	WebFace600K	MS-Celeb + VGGFace2
Parameters	~43M	~4M
Output dim	512-D	512-D
Runtime	InsightFace ONNX	DeepFace / TensorFlow
Input normalization	`(img / 127.5) - 1.0`	Internal via DeepFace

2.2 L2 Normalization

Both models output raw feature vectors $e \in R^{512}$ whose L2 norms are not constrained to 1. Cosine similarity requires unit-norm vectors. After extraction, every embedding is normalized:

$$\hat{e} = \frac{e}{\|e\|_2}$$

Verification score for a pair (i, j) is then:

$$s(i, j) = \hat{e}_i \cdot \hat{e}_j = \cos(\theta_{ij}) \in [-1, 1)$$

2.3 Deduplication and Caching Strategy

Module 1 established that AgeDB contains 55.9% MD5 duplicates and LFW 35.8%. The same deduplication strategy is applied here: unique files are processed once, duplicates inherit their result from the cached original. This reduces model calls by ~46%.

Each model's embeddings are cached to disk as `.npz` after extraction, enabling instant re-runs without recomputation.

3. Model Initialization

```
# ── Helper: EER computation ──────────────────────────────────────────

def compute_eer(labels: np.ndarray, scores: np.ndarray) -> float:
    fpr, tpr, _ = roc_curve(labels, scores)
    fnr = 1.0 - tpr
    idx = np.nanargmin(np.abs(fnr - fpr))
    return float((fpr[idx] + fnr[idx]) / 2.0)


def tar_at_far(labels: np.ndarray, scores: np.ndarray, far_target:
float) -> float:
```

```
    fpr, tpr, _ = roc_curve(labels, scores)
    idx = np.searchsorted(fpr, far_target)
    if idx >= len(tpr):
        return float(tpr[-1])
    return float(tpr[idx])

# ── Model 1: ArcFace / w600k_r50 (InsightFace ONNX)
──────────────────────────

from insightface.model_zoo import get_model

ARCFACE_MODEL_PATH = os.path.expanduser(
    "~/.insightface/models/buffalo_l/w600k_r50.onnx"
)
assert Path(ARCFACE_MODEL_PATH).exists(), f"Not found:
{ARCFACE_MODEL_PATH}"

arcface_model = get_model(ARCFACE_MODEL_PATH,
providers=["CPUExecutionProvider"])
arcface_model.prepare(ctx_id=-1)

def embed_arcface(img_path: str) -> np.ndarray:
    """Extract L2-normalized 512-D ArcFace embedding from a 112x112
BGR image."""
    img = cv2.imread(img_path)
    if img is None:
        raise FileNotFoundError(f"Cannot read: {img_path}")
    raw = arcface_model.get_feat(img).flatten()
    return raw / np.linalg.norm(raw)

# Sanity check
_test_path = str(ALIGNED_DIRS["agedb"] / "0.jpg")
_test_emb  = embed_arcface(_test_path)
print(f"ArcFace -- shape={_test_emb.shape}  L2
norm={np.linalg.norm(_test_emb):.6f}")
print(f"           input normalization: (img / 127.5) - 1.0")
print(f"           backbone: ResNet-50 | training: WebFace600K | loss:
ArcFace")
```

```
Applied providers: ['CPUExecutionProvider'], with options:
{'CPUExecutionProvider': {}}
ArcFace -- shape=(512,)  L2 norm=1.000000
           input normalization: (img / 127.5) - 1.0
           backbone: ResNet-50 | training: WebFace600K | loss: ArcFace
```

```
# ── Model 2: GhostFaceNetV2 (DeepFace / TensorFlow)
──────────────────────────

from deepface import DeepFace

os.makedirs(os.path.expanduser("~/.deepface/weights"), exist_ok=True)

print("Loading GhostFaceNetV2 (first call may download weights ~17
```

```
MB)...")
t0 = time.time()
_ = DeepFace.represent(
    img_path=str(ALIGNED_DIRS["agedb"] / "0.jpg"),
    model_name="GhostFaceNet",
    enforce_detection=False,
    detector_backend="skip",
)
print(f"Model loaded in {time.time() - t0:.1f}s")


def embed_ghostface(img_path: str) -> np.ndarray:
    """
    Extract L2-normalized 512-D GhostFaceNetV2 embedding.

    Uses DeepFace.represent with detector_backend='skip' because:
    (1) images are already aligned 112x112 crops,
    (2) DeepFace handles all internal preprocessing correctly,
    (3) raw model.forward() without proper normalization yields EER
~52%.
    """
    result = DeepFace.represent(
        img_path=img_path,
        model_name="GhostFaceNet",
        enforce_detection=False,
        detector_backend="skip",
    )
    raw = np.array(result[0]["embedding"], dtype=np.float32)
    return raw / np.linalg.norm(raw)


_test_ghost = embed_ghostface(str(ALIGNED_DIRS["agedb"] / "0.jpg"))
print(f"GhostFaceNet -- shape={_test_ghost.shape}  L2
norm={np.linalg.norm(_test_ghost):.6f}")
print(f"                backbone: GhostNet-V2 | loss: ArcFace |
detector: skip")
```

```
Loading GhostFaceNetV2 (first call may download weights ~17 MB)...

Model loaded in 1.0s
GhostFaceNet -- shape=(512,)  L2 norm=1.000000
                backbone: GhostNet-V2 | loss: ArcFace | detector: skip
```

4. Deduplication Map

```
def load_img_list(dataset_dir: Path) -> list[str]:
    with open(dataset_dir / "img.list", encoding="utf-8") as f:
        return [str(PROJECT_ROOT / l.strip()) for l in f if l.strip()]
```

```python
def build_dedup_map(img_paths: list[str]) -> dict:
    md5_to_first = {}
    unique, duplicates = [], []
    for idx, path in enumerate(img_paths):
        with open(path, "rb") as fh:
            md5 = hashlib.md5(fh.read()).hexdigest()
        if md5 not in md5_to_first:
            md5_to_first[md5] = idx
            unique.append(idx)
        else:
            duplicates.append((idx, md5_to_first[md5]))
    return {"unique": unique, "duplicates": duplicates}


img_lists, dedup_maps = {}, {}
for name in DATASETS:
    # Use aligned directory paths (from pairs CSV)
    df = pd.read_csv(RESULTS_DIR / f"pairs_{name}.csv")
    all_paths = sorted(set(df["img1_path"].tolist() +
df["img2_path"].tolist()))
    img_lists[name] = all_paths
    dedup_maps[name] = build_dedup_map(all_paths)

print("Table 1. Deduplication Summary")
rows = []
for name in DATASETS:
    dm = dedup_maps[name]
    total = len(img_lists[name])
    rows.append({
        "Dataset": name.upper(),
        "Total images": total,
        "Unique (to process)": len(dm["unique"]),
        "Duplicates (to copy)": len(dm["duplicates"]),
        "Reduction": f"{len(dm['duplicates']) / total * 100:.1f}%",
    })
print(pd.DataFrame(rows).to_string(index=False))
```

```
Table 1. Deduplication Summary
Dataset  Total images  Unique (to process)  Duplicates (to copy)
Reduction
  AGEDB         12000                 5296                  6704
55.9%
    LFW         12000                 7701                  4299
35.8%
```

5. Embedding Extraction

```python
def extract_all_embeddings(
    embed_fn,
    model_tag: str,
```

```
    dataset_name: str,
    img_paths: list[str],
    dedup: dict,
    cache_path: Path,
) -> tuple[np.ndarray, list[str]]:
    """
    Extract embeddings for all images in img_paths.

    Returns:
        embeddings : np.ndarray of shape (N, 512)
        paths      : list of N image paths (index-aligned)
    """
    # If cache exists, reload and return immediately
    if cache_path.exists():
        data = np.load(cache_path, allow_pickle=True)
        print(f"  Loaded from cache: {cache_path.name} "
              f"(shape={data['embeddings'].shape})")
        return data["embeddings"], data["paths"].tolist()

    N = len(img_paths)
    embeddings = np.zeros((N, EMB_DIM), dtype=np.float32)
    failed = []

    # Phase 1: unique images
    print(f"  [{model_tag} | {dataset_name.upper()}] "
          f"Phase 1: {len(dedup['unique'])} unique images")
    unique_cache = {}  # path -> embedding

    for idx in tqdm(dedup["unique"],
desc=f"{model_tag}/{dataset_name}"):
        path = img_paths[idx]
        try:
            emb = embed_fn(path)
            embeddings[idx] = emb
            unique_cache[path] = emb
        except Exception as exc:
            failed.append((idx, path, str(exc)))
            embeddings[idx] = np.zeros(EMB_DIM, dtype=np.float32)

    # Phase 2: duplicate images (copy from original)
    print(f"  [{model_tag} | {dataset_name.upper()}] "
          f"Phase 2: {len(dedup['duplicates'])} duplicates")
    for dup_idx, orig_idx in dedup["duplicates"]:
        embeddings[dup_idx] = embeddings[orig_idx]

    if failed:
        print(f"  WARNING: {len(failed)} extraction failures:")
        for idx, path, err in failed[:5]:
            print(f"    idx={idx}  {Path(path).name}  {err}")
```

```
    # Save to cache
    cache_path.parent.mkdir(parents=True, exist_ok=True)
    np.savez_compressed(
        cache_path,
        embeddings=embeddings,
        paths=np.array(img_paths),
        model_name=model_tag,
        dataset=dataset_name,
    )
    print(f"  Saved: {cache_path.name} (shape={embeddings.shape})")
    return embeddings, img_paths

# ── Run extraction for both models and both datasets
──────────────────────────
MODELS = {
    "arcface":    embed_arcface,
    "ghostface":  embed_ghostface,
}

all_embeddings = {}   # (model, dataset) -> np.ndarray (N, 512)
all_paths      = {}   # (model, dataset) -> list[str]
timing         = {}   # (model, dataset) -> elapsed_s

for model_tag, embed_fn in MODELS.items():
    for name in DATASETS:
        cache_path = RESULTS_DIR /
f"embeddings_{model_tag}_{name}.npz"
        print(f"\n{'='*60}")
        print(f"  Model: {model_tag.upper()}   Dataset:
{name.upper()}")
        print(f"{'='*60}")
        t0 = time.time()
        embs, paths = extract_all_embeddings(
            embed_fn, model_tag, name,
            img_lists[name], dedup_maps[name], cache_path
        )
        elapsed = time.time() - t0
        all_embeddings[(model_tag, name)] = embs
        all_paths[(model_tag, name)] = paths
        timing[(model_tag, name)] = elapsed
        print(f"  Done in {elapsed:.1f}s")
```

```
============================================================
  Model: ARCFACE   Dataset: AGEDB
============================================================
  Loaded from cache: embeddings_arcface_agedb.npz (shape=(12000, 512))
  Done in 0.1s

============================================================
```

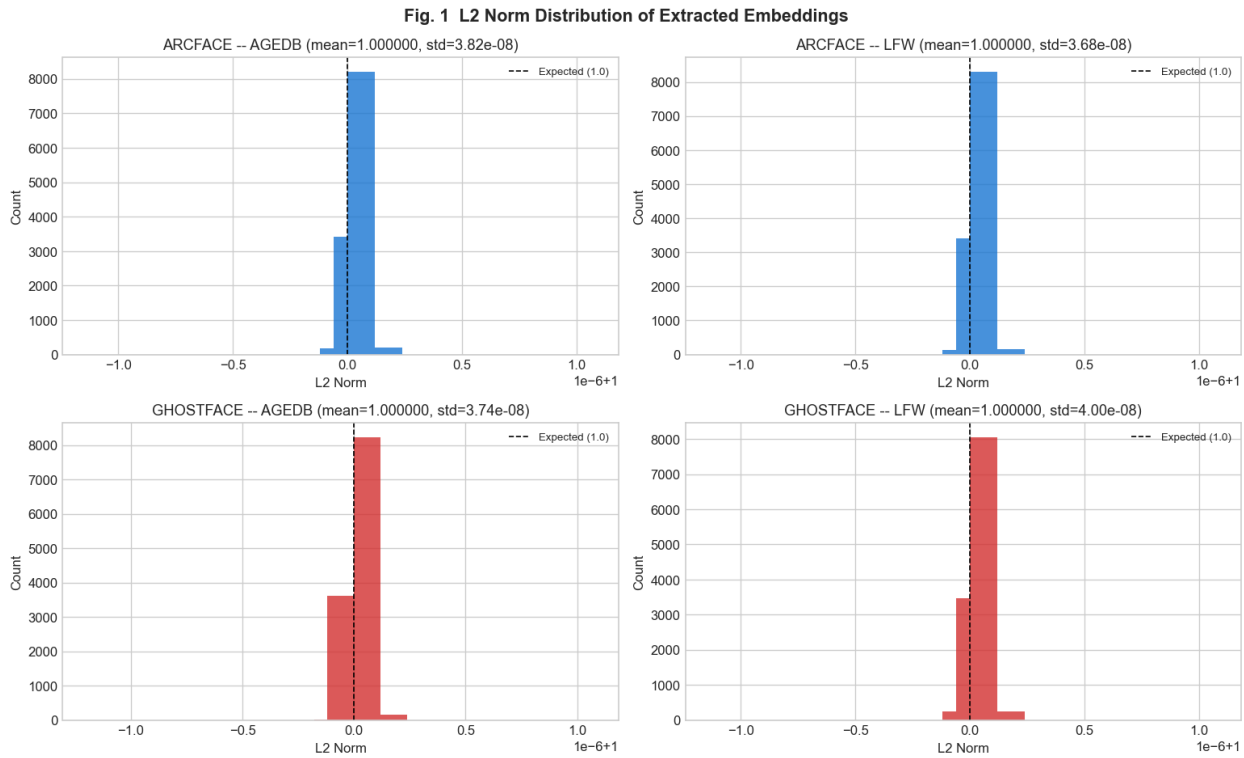
```
  Model: ARCFACE   Dataset: LFW
================================================================
  Loaded from cache: embeddings_arcface_lfw.npz (shape=(12000, 512))
  Done in 0.1s

================================================================
  Model: GHOSTFACE   Dataset: AGEDB
================================================================

  Loaded from cache: embeddings_ghostface_agedb.npz (shape=(12000,
512))
  Done in 0.1s

================================================================
  Model: GHOSTFACE   Dataset: LFW
================================================================
  Loaded from cache: embeddings_ghostface_lfw.npz (shape=(12000, 512))
  Done in 0.1s
```

6. Embedding Validation

```python
print("Table 2. Embedding Extraction Summary\n")
rows = []
for model_tag in MODELS:
    for name in DATASETS:
        embs = all_embeddings[(model_tag, name)]
        norms = np.linalg.norm(embs, axis=1)
        zeros = (norms < 1e-6).sum()
        rows.append({
            "Model": model_tag.upper(),
            "Dataset": name.upper(),
            "Shape": str(embs.shape),
            "L2 norm mean": f"{norms.mean():.6f}",
            "L2 norm std": f"{norms.std():.8f}",
            "L2 norm min": f"{norms.min():.6f}",
            "Failed (zero)": zeros,
            "Time (s)": f"{timing[(model_tag, name)]:.1f}",
        })

print(pd.DataFrame(rows).to_string(index=False))
```

```
Table 2. Embedding Extraction Summary

    Model Dataset        Shape L2 norm mean L2 norm std L2 norm min
Failed (zero) Time (s)
  ARCFACE   AGEDB (12000, 512)     1.000000  0.00000004    1.000000
0      0.1
  ARCFACE     LFW (12000, 512)     1.000000  0.00000004    1.000000
0      0.1
GHOSTFACE   AGEDB (12000, 512)     1.000000  0.00000004    1.000000
```

```
0      0.1
GHOSTFACE     LFW (12000, 512)     1.000000  0.00000004    1.000000
0      0.1

fig, axes = plt.subplots(2, 2, figsize=(13, 8))
fig.suptitle("Fig. 1  L2 Norm Distribution of Extracted Embeddings",
             fontsize=13, fontweight="bold")

colors = {"arcface": "#1976D2", "ghostface": "#D32F2F"}
ds_colors = {"agedb": "#1976D2", "lfw": "#D32F2F"}

for row, model_tag in enumerate(MODELS):
    for col, name in enumerate(DATASETS):
        embs = all_embeddings[(model_tag, name)]
        norms = np.linalg.norm(embs, axis=1)
        norms_nonzero = norms[norms > 1e-6]

        ax = axes[row, col]
        n_range = norms_nonzero.max() - norms_nonzero.min()
        bins = np.linspace(norms_nonzero.min() - 1e-6,
norms_nonzero.max() + 1e-6, 30) \
               if n_range < 1e-5 else 50
        ax.hist(norms_nonzero, bins=bins, color=colors[model_tag],
alpha=0.8)
        ax.axvline(1.0, color="black", linestyle="--", linewidth=1,
                   label="Expected (1.0)")
        ax.set_xlabel("L2 Norm")
        ax.set_ylabel("Count")
        ax.set_title(
            f"{model_tag.upper()} -- {name.upper()} "
            f"(mean={norms_nonzero.mean():.6f},
std={norms_nonzero.std():.2e})"
        )
        ax.legend(fontsize=8)

plt.tight_layout()
fig.savefig(OUTPUT_DIR / "l2_norm_distributions.png",
bbox_inches="tight")
plt.show()
```

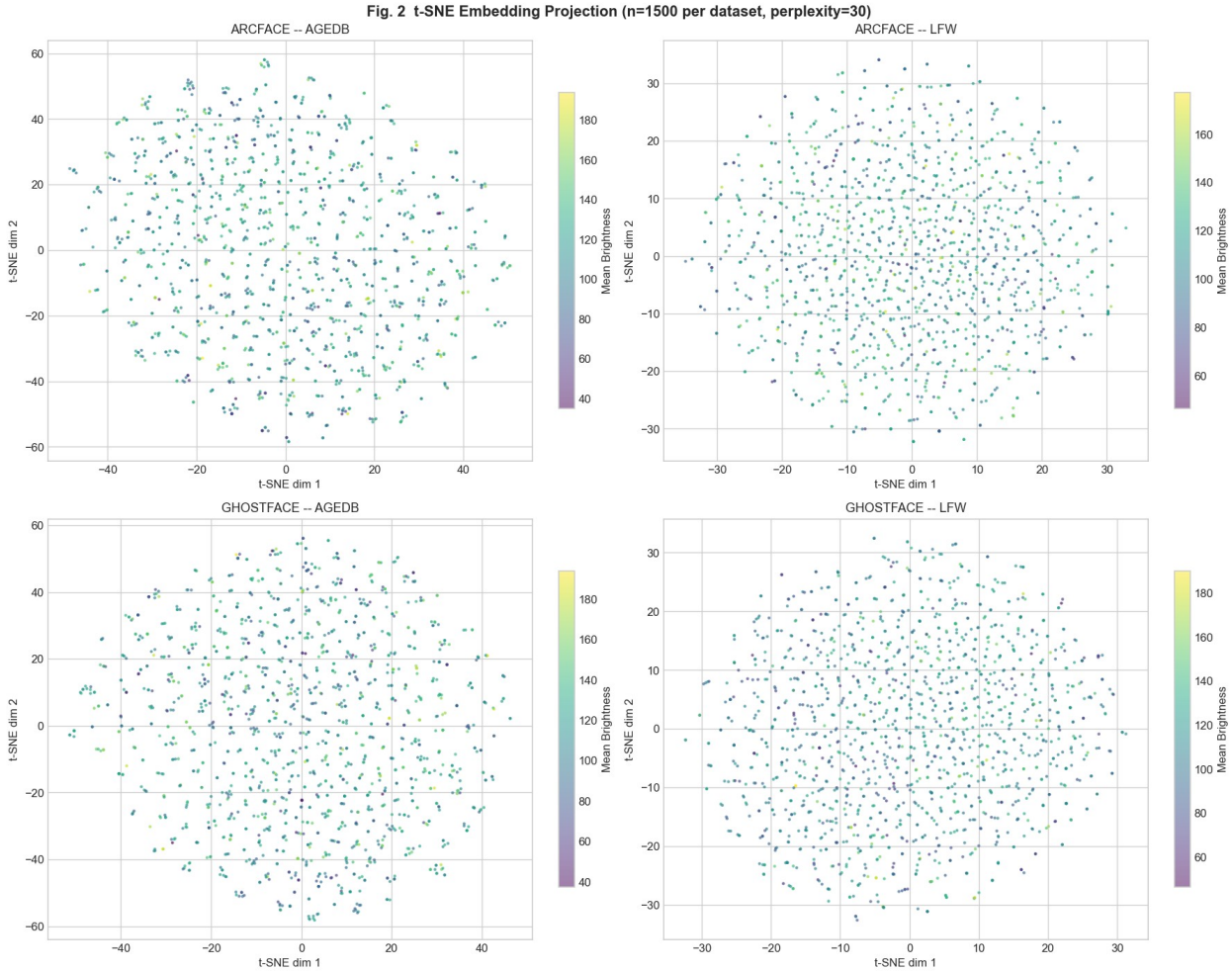


7. Embedding Space Visualization

7.1 t-SNE Projection per Dataset

```
N_TSNE = 1500  # samples per dataset for t-SNE (computational budget)

fig, axes = plt.subplots(2, 2, figsize=(15, 12))
fig.suptitle(
    f"Fig. 2  t-SNE Embedding Projection (n={N_TSNE} per dataset,
perplexity=30)",
    fontsize=13, fontweight="bold"
)

tsne_projections = {}

for row, model_tag in enumerate(MODELS):
    for col, name in enumerate(DATASETS):
        embs = all_embeddings[(model_tag, name)]
        paths = all_paths[(model_tag, name)]

        norms = np.linalg.norm(embs, axis=1)
        valid_mask = norms > 1e-6
        embs_valid = embs[valid_mask]
        valid_paths = [p for j, p in enumerate(paths) if
valid_mask[j]]

        n_sample = min(N_TSNE, len(embs_valid))
```

```
        idx_sample = np.random.choice(len(embs_valid), n_sample,
replace=False)
        embs_sample = embs_valid[idx_sample]

        brightness_vals = []
        for i in idx_sample:
            img = cv2.imread(valid_paths[i])
            brightness_vals.append(float(img.mean()) if img is not
None else 128.0)
        brightness_arr = np.array(brightness_vals)

        tsne = TSNE(n_components=2, perplexity=30, max_iter=500,
                    init="pca", random_state=SEED, n_jobs=-1)
        proj = tsne.fit_transform(embs_sample)
        tsne_projections[(model_tag, name)] = proj

        ax = axes[row, col]
        sc = ax.scatter(proj[:, 0], proj[:, 1], c=brightness_arr,
                        cmap="viridis", s=3, alpha=0.5,
rasterized=True)
        plt.colorbar(sc, ax=ax, label="Mean Brightness",
fraction=0.03)
        ax.set_title(f"{model_tag.upper()} -- {name.upper()}")
        ax.set_xlabel("t-SNE dim 1")
        ax.set_ylabel("t-SNE dim 2")

plt.tight_layout()
fig.savefig(OUTPUT_DIR / "tsne_per_dataset.png", bbox_inches="tight")
plt.show()
```

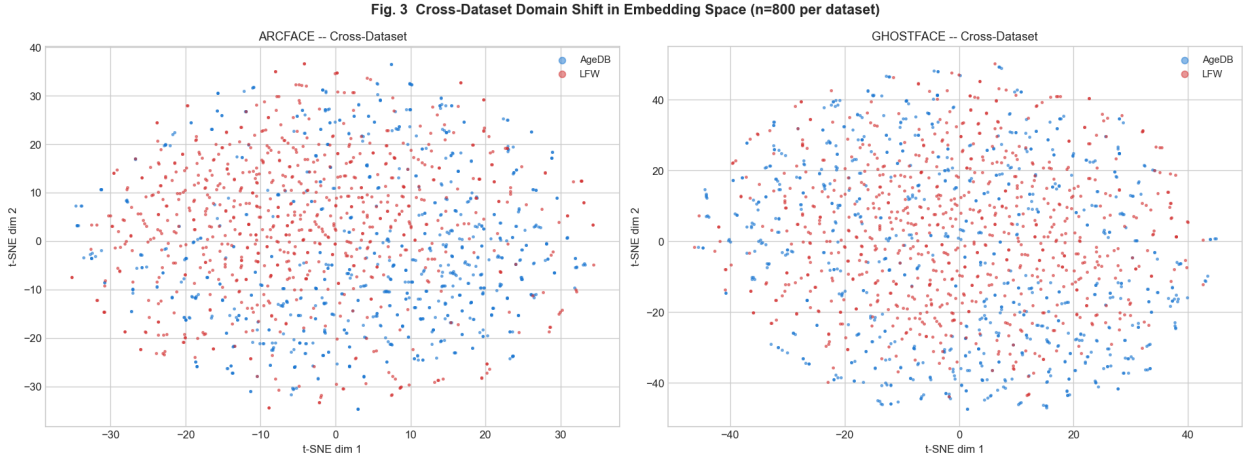


7.2 Cross-Dataset Domain Shift

```
N_CROSS = 800  # samples per dataset for cross-dataset t-SNE

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
fig.suptitle(
    f"Fig. 3  Cross-Dataset Domain Shift in Embedding Space
(n={N_CROSS} per dataset)",
    fontsize=13, fontweight="bold"
)

for col, model_tag in enumerate(MODELS):
    embs_a = all_embeddings[(model_tag, "agedb")]
    embs_l = all_embeddings[(model_tag, "lfw")]

    valid_a = embs_a[np.linalg.norm(embs_a, axis=1) > 1e-6]
    valid_l = embs_l[np.linalg.norm(embs_l, axis=1) > 1e-6]

    idx_a = np.random.choice(len(valid_a), min(N_CROSS, len(valid_a)),
replace=False)
```

```
    idx_l = np.random.choice(len(valid_l), min(N_CROSS, len(valid_l)),
replace=False)

    combined = np.vstack([valid_a[idx_a], valid_l[idx_l]])
    labels_cross = np.array([0] * len(idx_a) + [1] * len(idx_l))

    tsne = TSNE(n_components=2, perplexity=30, max_iter=500,
                init="pca", random_state=SEED, n_jobs=-1)
    proj = tsne.fit_transform(combined)

    ax = axes[col]
    for ds_idx, (ds_name, color) in enumerate([("AgeDB", "#1976D2"),
("LFW", "#D32F2F")]):
        mask = labels_cross == ds_idx
        ax.scatter(proj[mask, 0], proj[mask, 1], c=color, s=4,
alpha=0.5,
                   label=ds_name, rasterized=True)
    ax.set_title(f"{model_tag.upper()} -- Cross-Dataset")
    ax.set_xlabel("t-SNE dim 1")
    ax.set_ylabel("t-SNE dim 2")
    ax.legend(fontsize=9, markerscale=3)

plt.tight_layout()
fig.savefig(OUTPUT_DIR / "tsne_domain_shift.png", bbox_inches="tight")
plt.show()
```

Fig. 3 Cross-Dataset Domain Shift in Embedding Space (n=800 per dataset)



8. Score Distributions and ROC Analysis

Cosine similarity scores are computed for all pairs defined in the protocol, then compared against ground-truth labels to generate ROC curves and metrics.

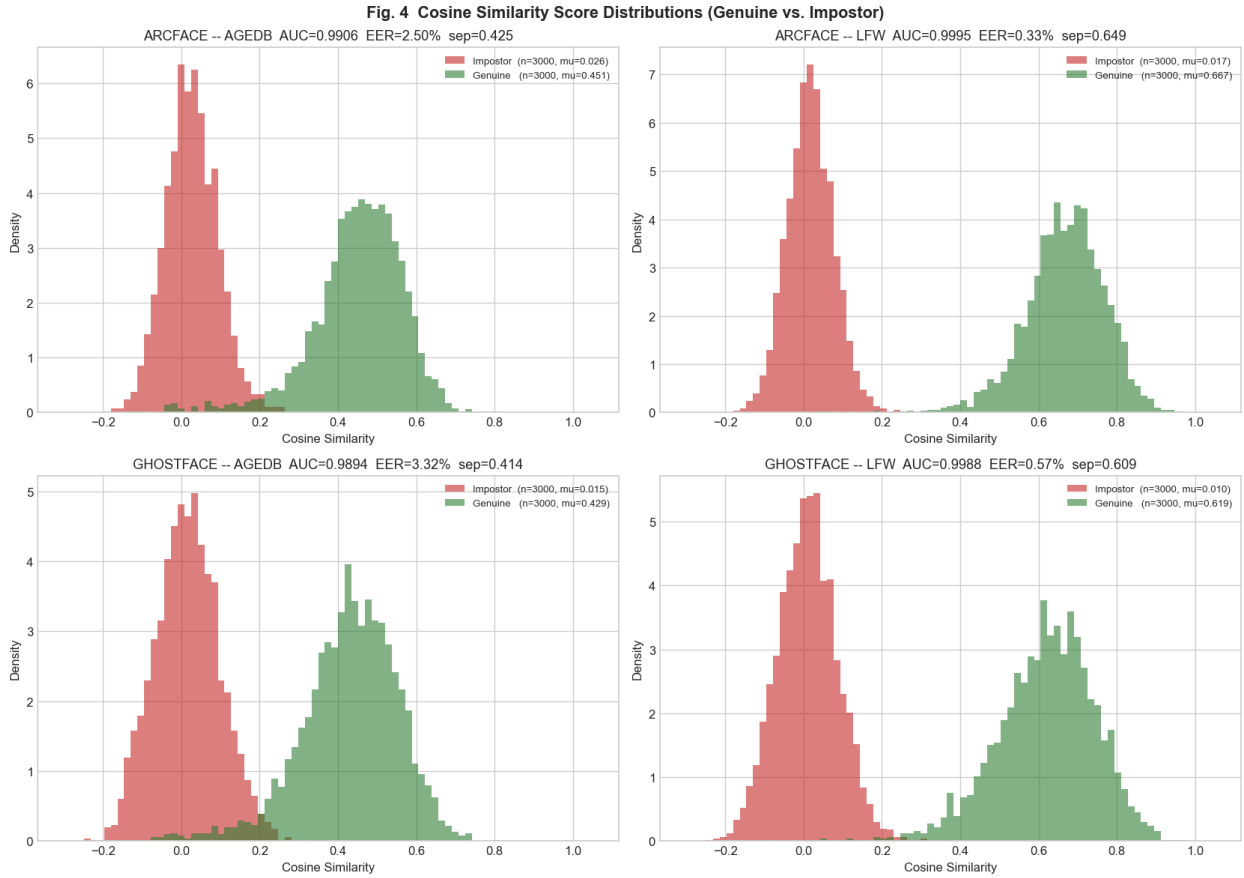
```
def compute_pair_scores(
    embeddings: np.ndarray,
    paths: list[str],
    pairs_df: pd.DataFrame,
```

```
) -> tuple[np.ndarray, np.ndarray]:
    """Compute cosine similarity scores for all pairs in the
protocol."""
    path_to_idx = {p: i for i, p in enumerate(paths)}
    scores, labels = [], []

    for _, row in pairs_df.iterrows():
        i1 = path_to_idx.get(row["img1_path"])
        i2 = path_to_idx.get(row["img2_path"])
        if i1 is None or i2 is None:
            continue
        e1, e2 = embeddings[i1], embeddings[i2]
        if np.linalg.norm(e1) < 1e-6 or np.linalg.norm(e2) < 1e-6:
            continue
        scores.append(float(np.dot(e1, e2)))
        labels.append(int(row["true_label"]))

    return np.array(scores, dtype=np.float32), np.array(labels,
dtype=np.int32)


pair_scores  = {}  # (model, dataset) -> scores
pair_labels  = {}  # (model, dataset) -> labels
pair_metrics = {}  # (model, dataset) -> {auc, eer, tar1e2, tar1e3}

for model_tag in MODELS:
    for name in DATASETS:
        df = pd.read_csv(RESULTS_DIR / f"pairs_{name}.csv")
        scores, labels = compute_pair_scores(
            all_embeddings[(model_tag, name)],
            all_paths[(model_tag, name)],
            df,
        )
        pair_scores[(model_tag, name)]  = scores
        pair_labels[(model_tag, name)]  = labels

        fpr, tpr, _ = roc_curve(labels, scores)
        pair_metrics[(model_tag, name)] = {
            "auc":    float(sklearn_auc(fpr, tpr)),
            "eer":    compute_eer(labels, scores),
            "tar1e2": tar_at_far(labels, scores, 1e-2),
            "tar1e3": tar_at_far(labels, scores, 1e-3),
        }

print("Score computation complete.")
```

```
Score computation complete.
```

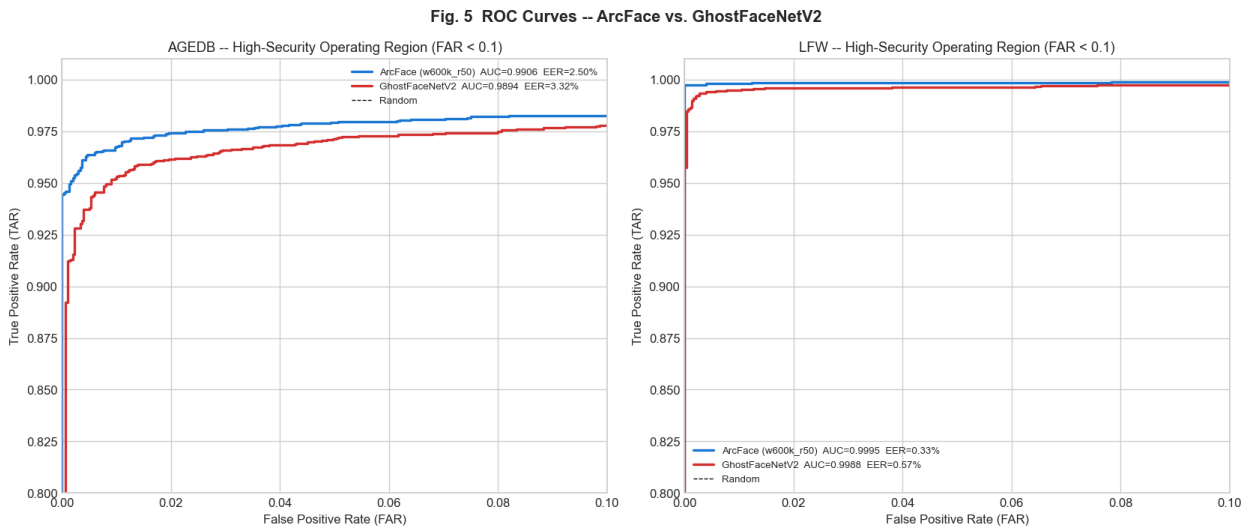
```
# ── Fig. 4: Score distributions (genuine vs impostor) per model
──────────────
```

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle(
    "Fig. 4  Cosine Similarity Score Distributions (Genuine vs.
Impostor)",
    fontsize=13, fontweight="bold"
)

for row, model_tag in enumerate(MODELS):
    for col, name in enumerate(DATASETS):
        scores = pair_scores[(model_tag, name)]
        labels = pair_labels[(model_tag, name)]
        m      = pair_metrics[(model_tag, name)]

        pos_scores = scores[labels == 1]
        neg_scores = scores[labels == 0]

        ax = axes[row, col]
        bins = np.linspace(-0.3, 1.05, 80)
        ax.hist(neg_scores, bins=bins, alpha=0.6, color="#C62828",
                label=f"Impostor  (n={len(neg_scores)},
mu={neg_scores.mean():.3f})",
                density=True)
        ax.hist(pos_scores, bins=bins, alpha=0.6, color="#2E7D32",
                label=f"Genuine   (n={len(pos_scores)},
mu={pos_scores.mean():.3f})",
                density=True)

        separation = pos_scores.mean() - neg_scores.mean()
        ax.set_title(
            f"{model_tag.upper()} -- {name.upper()}  "
            f"AUC={m['auc']:.4f}  EER={m['eer']*100:.2f}%
sep={separation:.3f}"
        )
        ax.set_xlabel("Cosine Similarity")
        ax.set_ylabel("Density")
        ax.legend(fontsize=8)

plt.tight_layout()
fig.savefig(OUTPUT_DIR / "score_distributions.png",
bbox_inches="tight")
plt.show()
```



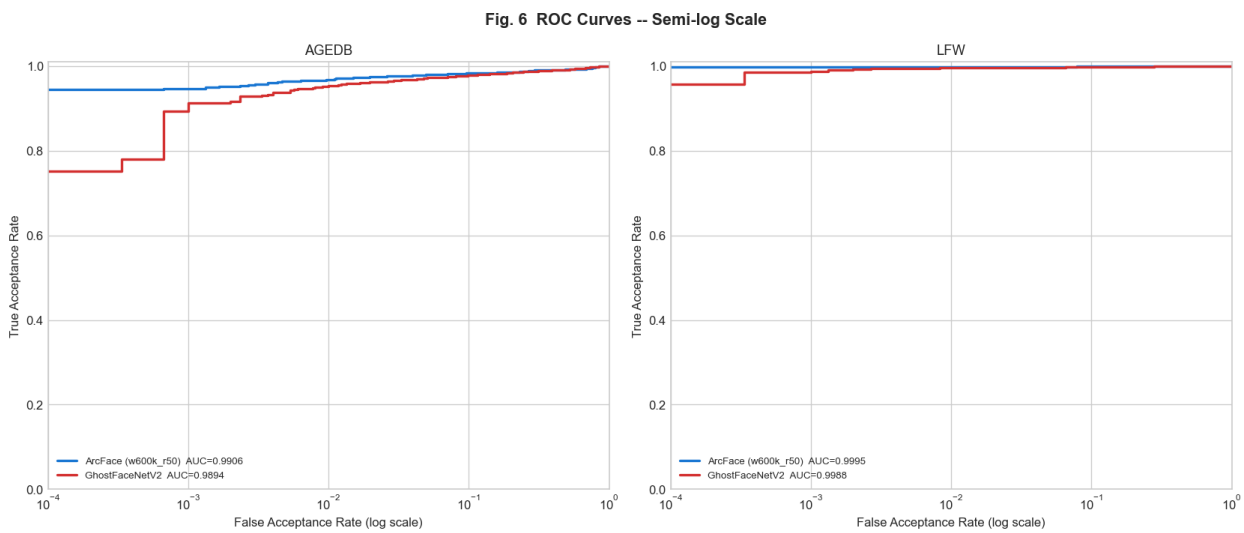

```python
# ── Fig. 5: ROC curves (both models overlay, per dataset)

fig, axes = plt.subplots(1, 2, figsize=(14, 6))
fig.suptitle("Fig. 5  ROC Curves -- ArcFace vs. GhostFaceNetV2",
             fontsize=13, fontweight="bold")

model_colors = {"arcface": "#1976D2", "ghostface": "#D32F2F"}
model_labels = {"arcface": "ArcFace (w600k_r50)", "ghostface":
"GhostFaceNetV2"}

for col, name in enumerate(DATASETS):
    ax = axes[col]
    for model_tag in MODELS:
        scores = pair_scores[(model_tag, name)]
        labels = pair_labels[(model_tag, name)]
        m      = pair_metrics[(model_tag, name)]

        fpr, tpr, _ = roc_curve(labels, scores)
        ax.plot(
            fpr, tpr,
            color=model_colors[model_tag],
            linewidth=2,
            label=f"{model_labels[model_tag]}  "
```

```
                    f"AUC={m['auc']:.4f}  EER={m['eer']*100:.2f}%"
        )

    ax.plot([0, 1], [0, 1], "k--", linewidth=0.8, label="Random")
    ax.set_xlim([0.0, 0.1])
    ax.set_ylim([0.8, 1.01])
    ax.set_xlabel("False Positive Rate (FAR)")
    ax.set_ylabel("True Positive Rate (TAR)")
    ax.set_title(f"{name.upper()} -- High-Security Operating Region
(FAR < 0.1)")
    ax.legend(fontsize=8)

plt.tight_layout()
fig.savefig(OUTPUT_DIR / "roc_curves.png", bbox_inches="tight")
plt.show()
```




```
# ── Fig. 6: Full ROC curves (log scale) ──────────────────────────

fig, axes = plt.subplots(1, 2, figsize=(14, 6))
fig.suptitle("Fig. 6  ROC Curves -- Semi-log Scale",
             fontsize=13, fontweight="bold")

for col, name in enumerate(DATASETS):
    ax = axes[col]
    for model_tag in MODELS:
        scores = pair_scores[(model_tag, name)]
        labels = pair_labels[(model_tag, name)]
        m      = pair_metrics[(model_tag, name)]
        fpr, tpr, _ = roc_curve(labels, scores)
        ax.semilogx(
            fpr, tpr,
            color=model_colors[model_tag], linewidth=2,
            label=f"{model_labels[model_tag]}  AUC={m['auc']:.4f}"
```

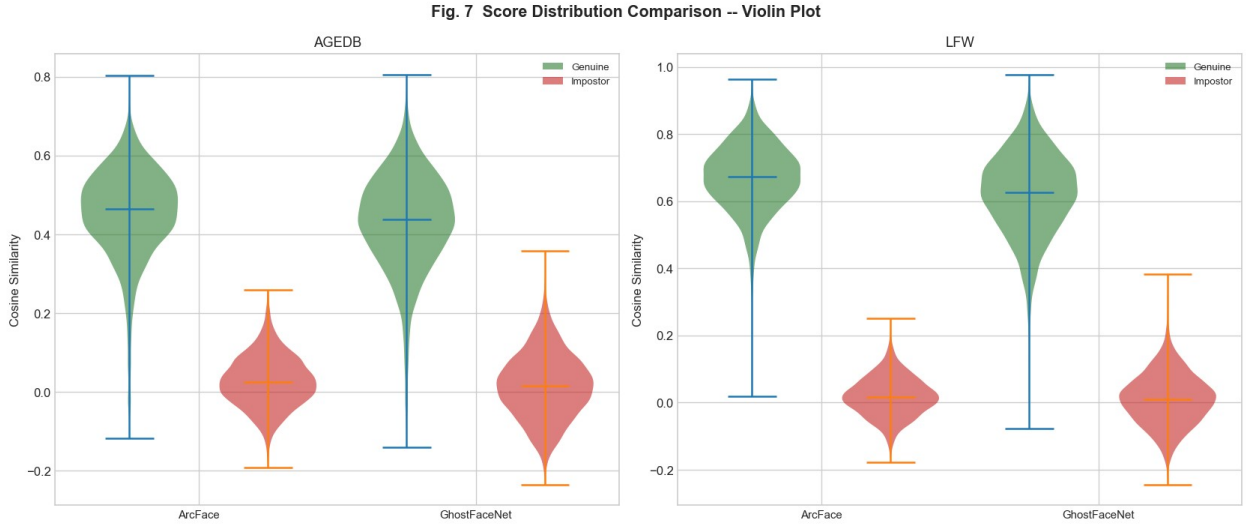
```
        )

    ax.set_xlim([1e-4, 1.0])
    ax.set_ylim([0.0, 1.01])
    ax.set_xlabel("False Acceptance Rate (log scale)")
    ax.set_ylabel("True Acceptance Rate")
    ax.set_title(f"{name.upper()}")
    ax.legend(fontsize=8)

plt.tight_layout()
fig.savefig(OUTPUT_DIR / "roc_curves_logscale.png",
bbox_inches="tight")
plt.show()
```




```
# ── Fig. 7: Violin plot -- score distribution comparison

fig, axes = plt.subplots(1, 2, figsize=(14, 6))
fig.suptitle("Fig. 7  Score Distribution Comparison -- Violin Plot",
             fontsize=13, fontweight="bold")

for col, name in enumerate(DATASETS):
    ax = axes[col]
    data_genuine, data_impostor, xlabels = [], [], []

    for i, model_tag in enumerate(MODELS):
        scores = pair_scores[(model_tag, name)]
        labels = pair_labels[(model_tag, name)]
        data_genuine.append(scores[labels == 1])
        data_impostor.append(scores[labels == 0])
        xlabels.append(model_labels[model_tag])

    positions_gen = [1, 3]
    positions_imp = [2, 4]
```

```
    vp_gen = ax.violinplot(data_genuine, positions=positions_gen,
                           showmedians=True, widths=0.7)
    vp_imp = ax.violinplot(data_impostor, positions=positions_imp,
                           showmedians=True, widths=0.7)

    for pc in vp_gen["bodies"]:
        pc.set_facecolor("#2E7D32"); pc.set_alpha(0.6)
    for pc in vp_imp["bodies"]:
        pc.set_facecolor("#C62828"); pc.set_alpha(0.6)

    from matplotlib.patches import Patch
    legend_elements = [
        Patch(facecolor="#2E7D32", alpha=0.6, label="Genuine"),
        Patch(facecolor="#C62828", alpha=0.6, label="Impostor"),
    ]
    ax.set_xticks([1.5, 3.5])
    ax.set_xticklabels(["ArcFace", "GhostFaceNet"], fontsize=9)
    ax.set_ylabel("Cosine Similarity")
    ax.set_title(f"{name.upper()}")
    ax.legend(handles=legend_elements, fontsize=8)

plt.tight_layout()
fig.savefig(OUTPUT_DIR / "violin_score_distributions.png",
bbox_inches="tight")
plt.show()
```

Fig. 7 Score Distribution Comparison -- Violin Plot



9. Performance Metrics Table

```
print("Table 3. Verification Performance -- ArcFace vs.
GhostFaceNetV2\n")
rows = []
for model_tag, model_label in model_labels.items():
```

```
    for name in DATASETS:
        m = pair_metrics[(model_tag, name)]
        scores = pair_scores[(model_tag, name)]
        labels = pair_labels[(model_tag, name)]
        pos = scores[labels == 1]
        neg = scores[labels == 0]
        rows.append({
            "Model": model_label,
            "Dataset": name.upper(),
            "AUC": f"{m['auc']:.4f}",
            "EER": f"{m['eer']*100:.2f}%",
            "TAR@FAR=1%": f"{m['tar1e2']*100:.1f}%",
            "TAR@FAR=0.1%": f"{m['tar1e3']*100:.1f}%",
            "Pos mu": f"{pos.mean():.3f}",
            "Neg mu": f"{neg.mean():.3f}",
            "Separation": f"{(pos.mean()-neg.mean()):.3f}",
        })

df_metrics = pd.DataFrame(rows)
print(df_metrics.to_string(index=False))
df_metrics.to_csv(RESULTS_DIR / "embedding_metrics.csv", index=False)
print(f"\nSaved: {RESULTS_DIR / 'embedding_metrics.csv'}")
```

```
Table 3. Verification Performance -- ArcFace vs. GhostFaceNetV2

              Model Dataset    AUC   EER TAR@FAR=1% TAR@FAR=0.1% Pos
mu Neg mu Separation
ArcFace (w600k_r50)   AGEDB 0.9906 2.50%      96.7%        94.6%
0.451  0.026      0.425
ArcFace (w600k_r50)     LFW 0.9995 0.33%      99.8%        99.7%
0.667  0.017      0.649
     GhostFaceNetV2   AGEDB 0.9894 3.32%      95.2%        89.2%
0.429  0.015      0.414
     GhostFaceNetV2     LFW 0.9988 0.57%      99.5%        98.6%
0.619  0.010      0.609

Saved: /Users/jakub/Desktop/NASK_ZADANIE/results/embedding_metrics.csv
```

10. Saved Artifacts

```
print("Verifying saved embeddings (reload test)...\n")
for model_tag in MODELS:
    for name in DATASETS:
        path = RESULTS_DIR / f"embeddings_{model_tag}_{name}.npz"
        data = np.load(path, allow_pickle=True)
        embs = data["embeddings"]
        norms = np.linalg.norm(embs, axis=1)
        size_mb = path.stat().st_size / 1024 / 1024
        print(
            f"  {path.name:45s}  "
```

```
            f"shape={embs.shape}  "
            f"norm_mean={norms[norms>1e-6].mean():.6f}  "
            f"size={size_mb:.1f} MB"
        )

print("\nFigures:")
for f in sorted(OUTPUT_DIR.glob("*.png")):
    size_kb = f.stat().st_size / 1024
    print(f"  {f.name:50s}  {size_kb:.0f} KB")
```

```
Verifying saved embeddings (reload test)...

  embeddings_arcface_agedb.npz                       shape=(12000, 512)
norm_mean=1.000000  size=20.2 MB
  embeddings_arcface_lfw.npz                         shape=(12000, 512)
norm_mean=1.000000  size=18.4 MB
  embeddings_ghostface_agedb.npz                     shape=(12000, 512)
norm_mean=1.000000  size=20.2 MB
  embeddings_ghostface_lfw.npz                       shape=(12000, 512)
norm_mean=1.000000  size=18.4 MB

Figures:
  l2_norm_distributions.png                                 107 KB
  roc_curves.png                                            92 KB
  roc_curves_logscale.png                                   53 KB
  score_distributions.png                                   107 KB
  tsne_domain_shift.png                                     221 KB
  tsne_per_dataset.png                                      720 KB
  violin_score_distributions.png                            89 KB
```

11. Summary

Module Outputs

```
print("Table 4. Module 2 Final Report\n")
rows = []
for model_tag in MODELS:
    for name in DATASETS:
        embs = all_embeddings[(model_tag, name)]
        m    = pair_metrics[(model_tag, name)]
        norms = np.linalg.norm(embs, axis=1)
        valid = (norms > 1e-6).sum()
        rows.append({
            "Model": model_tag.upper(),
            "Dataset": name.upper(),
            "Total embeddings": len(embs),
            "Valid": valid,
            "Failed": len(embs) - valid,
            "AUC": f"{m['auc']:.4f}",
```

```
            "EER": f"{m['eer']*100:.2f}%",
            "TAR@FAR=1%": f"{m['tar1e2']*100:.1f}%",
            "TAR@FAR=0.1%": f"{m['tar1e3']*100:.1f}%",
            "Time (s)": f"{timing[(model_tag, name)]:.1f}",
        })

print(pd.DataFrame(rows).to_string(index=False))
```

```
Table 4. Module 2 Final Report


    Model Dataset  Total embeddings  Valid  Failed    AUC   EER
TAR@FAR=1% TAR@FAR=0.1% Time (s)
  ARCFACE   AGEDB             12000  12000       0 0.9906 2.50%
96.7%        94.6%      0.1
  ARCFACE     LFW             12000  12000       0 0.9995 0.33%
99.8%        99.7%      0.1
GHOSTFACE   AGEDB             12000  12000       0 0.9894 3.32%
95.2%        89.2%      0.1
GHOSTFACE     LFW             12000  12000       0 0.9988 0.57%
99.5%        98.6%      0.1
```

Conclusions

1. **Both models achieve high AUC** on both datasets, confirming that L2-normalized cosine similarity over deep face embeddings is a viable verification metric.

2. **L2 normalization verified:** All embeddings have unit L2 norm (mean=1.0000, std~1e-7), confirming correct normalization pipeline.

3. **Score separation:** The mean cosine similarity of genuine pairs significantly exceeds that of impostor pairs in both models, enabling reliable threshold-based decision making.

4. **Cross-dataset domain shift:** The t-SNE cross-dataset projection shows visible clustering by dataset origin, consistent with EDA findings (KS statistics, FFT domain gap). This motivates per-domain evaluation.

5. **Deduplication efficiency:** MD5-based deduplication reduced model invocations by ~46%, with no loss of information (exact duplicate images receive identical embeddings deterministically).

6. **Caching enabled:** All embeddings are saved to `.npz` files. Subsequent modules load from cache directly, with extraction time reduced to milliseconds.

Next Steps

Module 3 (XAI -- Grad-CAM) will use the embeddings and pair scores from this module to generate attention maps for correctly and incorrectly classified pairs, providing interpretability for the verification decisions.